Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Mid-Term Examination, July-December-2021

Program: B. Sc. Engg.(CSE)
Session : **2018-19**

Course Code : CIT-323 | Course Title : Simulation and Modeling

Full Time: 1 Hour | Full Marks: 15

Answer all the questions

1. A drunkard moved from a point to a destination and takes step in four directions, backward, forward, right, and left. The probabilities associated with these are 10%, 50%, 15% and 25% respectively. The distances covered to backward, forward, right, and left are 30 cm, 80 cm, 60 cm and 40 cm respectively. Simulate the walk for 20 steps and find the location at end of 20 steps while starting point is (0, 0) on the X-Y scales. [USE Random Number 1 to 20 from table] 05

2. An ammunition depot, of rectangular shape measuring 500 m of length and 350m of width is under attack by a squadron of bombers. In each sortie 10 bombers drop bombs, one each, on the ammunition depot. If the bomb lands anywhere in the marked area, it is a hit, otherwise it is a miss. All the bombers aim at the center of the area. The point of strike is assumed to be normally distributed around the aiming point with a standard deviation of 500 m in x-direction and 350m in the y-direction. Simulate operation of bombing operation for 20 strikes and determine the percentage number of strikes, Which are on the target. [USE Random Number 21 to 40 for X coordinate and Random Number 41 to 60 for Y coordinate ,from table] 05

3. Define Monte-Carlo method. Solve the integrals $I = \int_2^5 x^3 dx$ by monte-carlo method. [USE Random Number 61 to 70 for X coordinate and Random Number 71 to 80 for Y coordinate, from table] 05

$x = 0.1$ (D)
$y = 190$ (P)
5 RN x RN y . x^3 65 RN

(60, 0)

-160, 660
$\frac{59}{9} - \frac{1}{9}$

484000
156.25 +

Patuakhali Science and Technology University (PSTU)

Department of Computer Science and Information Technology(CIT)

Mid-Term Examination, July-December -2020
Program: B. Sc.Engg. in CSE
Session : 2017-18

Course Code : CIT-323 | Course Title : Simulation and Modeling

Full Time: 1hour | Full Marks: 15

Answer all the questions

1. Define system simulation. Describe different types of simulation models. 03

2. What are the advantages of simulation? Describe about the applied area of simulation. 02

3. A drunkard moved from a point to a destination and takes step in three directions, forward, backward, left and right. The probabilities associated with these are 40%, 15%, 22% and 23% respectively. The distances covered to forward, left and right are 75 cm, 45 cm, 60 cm and 60 cm respectively. Simulate the walk for 25 steps and find the location at end of 25 steps while starting point is (0, 0) on the X-Y scales. [USE RN 67, 23, 01, 62, 84, 56, 77, 76, 98, 82, 45, 87, 29, 68, 21, 15, 36, 00, 89, 43, 15, 56, 89, 43, 32] 05

4. Define Monte-Carlo method. Solve the integrals $I = \int_2^5 x^3 dx$ by monte-carlo method. [USE Random Number 22,25,18,45,15,27,48,43,40,47 for X coordinate and Random Number .57, .18, .00, .90, .05, .77, .66, .10, .76, .42 for Y coordinate, from table] 05.

0 1 2 3

00 - 99 = F

40 - 59 = B 0 - 90

55 - 77 = R

78 - 99 = L

Simulation

Patuakhali Science and Technology University (PSTU)

Department of Computer Science and Information Technology(CIT)

Mid-Term Examination, July-December -2018
Program: B. Sc.Engg. in CSE
Session : 2015-16

Course Code : CIT-323

Course Title : Simulation and Modeling

Full Time: 1hour

Full Marks: 15

Answer all the questions

1. Define system simulation. Describe different types of simulation models. 03

2. What are the advantages of simulation? Describe about the applied area of simulation. 02

3. A drunkard moved from a point to a destination and takes step in ~~three~~ four directions, forward, backward, left and right. The probabilities associated with these are 40%,15%, 22% and 23% respectively. The distances covered to forward, left and right are 75 cm, 45 cm, 60 cm and 60 cm respectively. Simulate the walk for 25 steps and find the location at end of 25 steps while starting point is (0, 0) on the X-Y scales. [USE RN 67, 23, 01, 62, 84, 56, 77, 76, 98, 82, 45, 87, 29, 68, 21, 15, 36, 00, 89, 43, 15, 56, 89, 43, 32] 05

4. Consider a chemical reaction of two gas H_2 *and* O_2 react to produce H_2O. The rate of the reaction is depending upon the ratio in which H_2 *and* O_2 are mixed and some other factor like temperature, pressure and humidity which remains constant during the reaction. There will also happen backward reaction where H_2O decomposes back into H_2 *and* O_2. If C_1, C_2 *and* C_3 are the amount of H_2, O_2 *and* H_2O at any instant of time t, the rate of increase of H_2, O_2 *and* H_2O are given by the following differential equations: 05

$$\frac{dC_1}{dt} = K_2C_3 - K_1C_1C_2$$
$$\frac{dC_2}{dt} = K_2C_3 - K_1C_1C_2$$
$$\frac{dC_3}{dt} = 2K_1C_1C_2 - 2K_2C_3$$

Where K_1 *and* K_2 are constants. If K_1 =0.025; $K_2 = 0.01$ and at time t=0, $C_1 = 50.00\,m^3$ $C_2 = 25.00\,m^3$ $C_3 = 0.00\,m^3$ $\Delta t = 0.2\,minute$. Simulate the experiment for 3 minute to determine the amount of H_2, O_2 *and* H_2O.

Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Final Examination, July-December-2021

Program: B. Sc. Engg.(CSE)
Session : **2018-19**

Course Code : CIT-323 | Course Title : Simulation and Modeling

Full Time: 3 Hours | Full Marks: 70

[Figures in the right margin indicate full marks]
Answer any 5 of the following questions.

1. a. Define the following terminologies: i) System ii) Model iii) Simulation iv) Analytical solution. Demonstrate different ways in which a system might be studied. 04

   b. What are the steps of simulation for a single-server queueing system? Describe the problem statement and intuitive explanation of a single-server queueing system. 10

2. a. What is random variable? Compute variance and covariance based on a set of random variates. Why stochastic process is used to simulate the output data? Illustrate correlation function of the process of delays in queue D1, D2, ...for the M/M/1 queue. 07

   b. Explain hypothesis testing including null and alternative hypothesis for mean. What is the difference type I error and type II error? Explain and illustrate the strong law of large number. 07

3. a. Write the properties of location, scale, and shape parameters of continuous distributions. What is inverse transform function? Let X have the exponential distribution with mean β, Generate the desired random variate using the inverse transform function. 07

   b. What is Monte Carlo Method? Write down the difference between monte-carlo and stochastic simulation. 02

   c. Define Monte-Carlo method. Solve the integrals$I = \int_1^5 \frac{x^4}{3} dx$ by monte-carlo method. [USE Random Number 48, 36, 44, 41, 46, 18, 32, 41, 23, 23 for X coordinate and Random Number .52, .18, .30, .88. .21, .25, .57, .82, .9, .63 for Y coordinate, you have a right to do scaling, if required] 05

4. a. Define computer simulation models. Write down the qualities of an efficient random number generator. 02

   b. A machine center has three identical bearings, which fail according to the following probability distribution in table A. The present maintenance policy is to change a bearing as and when it fails. When bearing fails machine center stops, a technician is called to replace the failed bearing with a new one. The time between the failure of the bearing and reporting of the technician (delay time) is random and is distributed as table B. 10

Table: A	
Bearing Life(Hours)	***Probability***
1200	0.10
1400	0.25
1600	0.30
1800	0.25
2000	0.10

Table: B	
Delay Time(Minutes)	***Probability***
2	0.3
5	0.5
8	0.2

It takes 15 minutes to change one bearing, 25 minutes to change two bearing and 35 minutes to change all three bearings. Downtime of the system costs $5 per minutes, direct on job costs of the technician is $30 per hour and the cost of bearing is $30. The maintenance department is interested in evaluating an alternative policy of replacing all the three bearings, whenever a bearing fails. Simulate the present and proposed policy for 10000 hours and choose the better policy.

   c. What is normally distributed random numbers? Describe the properties and applications of normally distributed random numbers. 02

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5. a. Define the simulation of continuous system. Describe the disadvantages of analog simulation. 0

b. There are a number of moving objects, which chase each other in a serial order like A chases B, B chases C, and C chases D etc. Each object moves towards its target unmindful of the fact that it itself is being targeted by some other object. Depending upon the original location, and speed of motion of the object, each object will take its own time to hit its target. As soon as hit occurs, the chase ends. The initial location of the objects A, B, C, and D are (0,0), (0,10), (0,20), (0,30) respectively and their velocities are 30km/hr, 25km/hr, 20 km/hr and 15km/hr respectively. Object D moves towards a fixed at (30,50), while C moves always in a direction pointing towards D, the object B moves always pointing towards C and object A always pointing towards B. It can be assumed that when the distance between two objects less than 0.005 hit is taken to occur. Identify, which object will hit the target at first and in what time? 10

c. Why the random numbers generated by computer are called pseudo random numbers? Discuss the pitfalls of congruence method of generating random numbers. 02

6 a. Define random variables. What are the purposes of testing the uniformity and independence of random values? 02

b. i. A sequence of 100 random numbers is given below. Use Chi-Square test with $\alpha = 0.05$ to test these numbers are uniformly distributed. 12

21	81	92	23	96	20	68	57	79	84
82	62	12	08	92	83	74	85	60	49
48	37	65	74	22	11	28	10	55	82
72	95	08	85	79	95	86	11	16	52
70	55	50	87	67	51	72	38	29	62
71	12	07	75	56	34	40	67	24	86
18	82	41	29	63	06	84	01	20	06
06	33	14	79	25	65	57	47	74	68
54	35	81	07	88	96	70	85	29	13
12	91	26	57	30	22	90	03	13	31

ii. Define autocorrelation. Test the autocorrelation of the given numbers (6(b)i) by employing the chi-squared test with 99% confidence level.

N.B: You can use chi-squared table below as required.

Table: Chi-square Distribution

d.f.	.995	.99	.975	.95	.9	.1	.05	.025	.01
1	0.00	0.00	0.00	0.00	0.02	2.71	3.84	5.02	6.63
2	0.01	0.02	0.05	0.10	0.21	4.61	5.99	7.38	9.21
3	0.07	0.11	0.22	0.35	0.58	6.25	7.81	9.35	11.34
4	0.21	0.30	0.48	0.71	1.06	7.78	9.49	11.14	13.28
5	0.41	0.55	0.83	1.15	1.61	9.24	11.07	12.83	15.09
6	0.68	0.87	1.24	1.64	2.20	10.64	12.59	14.45	16.81
7	0.99	1.24	1.69	2.17	2.83	12.02	14.07	16.01	18.48
8	1.34	1.65	2.18	2.73	3.49	13.36	15.51	17.53	20.09
9	1.73	2.09	2.70	3.33	4.17	14.68	16.92	19.02	21.67
10	2.16	2.56	3.25	3.94	4.87	15.99	18.31	20.48	23.21
11	2.60	3.05	3.82	4.57	5.58	17.28	19.68	21.92	24.72
12	3.07	3.57	4.40	5.23	6.30	18.55	21.03	23.34	26.22
13	3.57	4.11	5.01	5.89	7.04	19.81	22.36	24.74	27.69
14	4.07	4.66	5.63	6.57	7.79	21.06	23.68	26.12	29.14
15	4.60	5.23	6.26	7.26	8.55	22.31	25.00	27.49	30.58
16	5.14	5.81	6.91	7.96	9.31	23.54	26.30	28.85	32.00
17	5.70	6.41	7.56	8.67	10.09	24.77	27.59	30.19	33.41
18	6.26	7.01	8.23	9.39	10.86	25.99	28.87	31.53	34.81
19	6.84	7.63	8.91	10.12	11.65	27.20	30.14	32.85	36.19
20	7.43	8.26	9.59	10.85	12.44	28.41	31.41	34.17	37.57

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Patuakhali Science and Technology University

Faculty of Computer Science and Engineering

Dept. of Computer Science and Information Technology

6th Semester (Level-3, Semester-II), **Final Examination of B.Sc. Engg. CSE**

July-December/2020, Session: 2017-18

Course Code: CIT-323 Course Title: Simulation and Modeling

Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks] Answer any 5 of the following questions

1. (a). Define computer simulation models. Describe the steps in simulation study. 02

(b). A drunkard moved from a point to a destination and takes step in **four** directions, backward, forward, right, and left. The probabilities associated with these are 10%, 50%, 15% and 25% respectively. The distances covered to backward, forward, right, and left are 30 cm, 80 cm, 60 cm and 40 cm respectively. Simulate the walk for 20 steps and find the location at end of 20 steps while starting point is (0, 0) on the X-Y scales.[Use Random Number; 6, 2, 0, 6, 2, 8, 5, 7, 7, 9, 8, 4, 8, 2, 6, 2, 1, 3, 0, 8, 4] 06

(c). An ammunition depot, of rectangular shape measuring 500 m of length and 350m of width is under attack by a squadron of bombers. In each sortie 10 bombers drop bombs, one each, on the ammunition depot. If the bomb lands anywhere in the marked area, it is a hit, otherwise it is a miss. All the bombers aim at the center of the area. The point of strike is assumed to be normally distributed around the aiming point with a standard deviation of 500 m in x-direction and 350m in the y-direction. Simulate operation of bombing operation for 20 strikes and determine the percentage number of strikes, Which are on the target. Use Normal Random Number: 06

For X direction – [.23, -1.16, 0.39, -1.90, -0.78, -0.02, -0.40, -0.66, 1.14, 0.07, 0.53, 0.03, 1.19,0.11, 0.16, -0.82, 0.42, -0.89, 0.49, -0.21]

For Y direction- [0.24, -0.02, 0.64, -1.04, 0.68, -0.47, -0.75, -0.44, 1.21, -0.08, -1.56, 1.49, -1.19, -1.86, 1.21, 0.75, -1.50, 0.19, -1.44, 0.65]

2. (a). Describe the iterative process of verification and validation of simulation models. 02

(b). Define the following terminologies: (i) Arrival rate (ii) Service rate 02

(c). In a pure pursuit, there is a target bombers which moves along a predetermine path (given on table below) and there is a pursuer aircraft who follows the target bombers, redirecting itself towards the target bombers at fixed intervals of time. 10

Target bombers moving path:

Time,t	0	1	2	3	4	5	6	7	8
xb(t)	160	161	167	172	181	189	199	215	239
yb(t)	0	4	9	11	14	24	34	41	52

Time,t	9	10	11	12	13	14	15		
xb(t)	245	254	265	271	289	293	300		
yb(t)	45	41	36	39	34	28	12		

The fighter aircraft following the enemy bombers with speed at 30KM/min to destroy it. The fighter aircraft is at position *xf, yf* (100, 50) when it sights the bomber that is at time t=0, the time at which the pursuit begins. The fighter corrects its direction after a fixed interval of one minute, so as to point towards a bomber and shoots the bomber by firing a missile as soon as it is within a distance of 12 KM. The

pursuit ends. In case the bomber is not shot within 10 minutes of the pursuit, the pursuit is abandoned. Simulate the problem to determine the pursuit is end/abandoned.

3. (a) Define distributed simulation. Write down the difference between monte-carlo and stochastic simulation. 4

(b). Consider a chemical reaction of two gas H_2 *and* O_2 react to produce H_2O. The rate of the reaction is depending upon the ratio in which H_2 *and* O_2 are mixed and some other factor like temperature, pressure and humidity which remains constant during the reaction. There will also happen backward reaction where H_2O decomposes back into H_2 *and* O_2. If C_1, C_2 *and* C_3 are the amount of H_2, O_2 *and* H_2O at any instant of time t, the rate of increase of H_2, O_2 *and* H_2O are given by the following differential equations: 5

$$\frac{dC_1}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_2}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_3}{dt} - 2K_1C_1C_2 - 2K_2C_3$$

Where, K_1 *and* K_2 are constants. If K_1=0.025; $K_1 = 0.01$ and at time t=0, $C_1 = 50.00\ m^3$ $C_2 = 25.00\ m^3$ $C_3 = 0.00\ m^3$ $\Delta t = 0.2\ minute$. Simulate the experiment for 3 minute to determine the amount of H_2, O_2 *and* H_2O.

(c). A sequence of 10,000 five digit random numbers has been generated, and an analysis of numbers indicate that there 3075 numbers having five different digits,4935 having a pair,1135 having two pairs, 695 having three of a kind,105 having full house, 54 having four of a kind and one having all five of a kind. Use Poker test to determine if these random numbers are independent at α = 0.01. Note that for α= 0.01 and N=06 the critical value is 16.8. 5

4. (a) What is normally distributed random numbers? Describe the application of normally distributed random numbers. 3

(b) The following sequence of random numbers have been generated 0.037, 0.55, 0.71, 0.97, 0.65, 0.29, 0.84, 0.78, 0.23,0.17; Use Kolmogorov-Smirnov test with α= 0.05 to determine, If these numbers are uniformly distributed over the interval 0 to 1. Note that for α= 0.05 and N=10 the critical value is 0.410. 5

(c). Define analog simulation. Describe the simulation of an exterior ballistics with an example. 6

5. (a) What is a stochastic process? Write the probability mass function of Poisson distribution. List several applications of Poisson distribution. 7

(b) Write the density function of the normal curve. Generate ten random observations from Normal distribution with a mean 15 and standard deviation 5. 7

6. (a) How to address the COVID-19 scenario using Exponential distribution. Illustrate the Exponential curves on the following scenarios: 7

(i) Exponential growth and Exponential decay.

(ii) Infection begins to taper off when herd immunity is reached.

(iii) Confirmed cases will rise again if restrictions are lifted too soon.

(b) What is the application of SIR model? Illustrate and state the general SIR schematic. Show and explain the maximum number of infected, I_{max} and solution curves. 7

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ for all real number x

Patuakhali Science and Technology University

6th semester (Level-3, Semester-II) Final Examination of B.Sc. Engg. (CSE), July-December 2019, Session 2016-17

Course Code: CIT-323 Course Title: Simulation and Modeling

Credit Hours: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions.

1. a) Define Simulation. Describe the applications of simulation in computer science. 2

b) A drunkard moved from a point to a destination and takes step in four directions, forward, backward, to left and right. The probabilities associated with these are 40%, 10%,25% and 25%. The distances covered in forward, backward, to left and right are 75 cm,45 cm,60 cm and 60 cm respectively. Simulate the walk for 20 steps and fond the location at end of 20 steps while starting point is (0,0) on the X-Y scales. [Use Random Number: 6,2,0,6,2,8,5,7,7,9,8,4,8,2,6,2,1,3,0,8,4] 5

c) What is real time simulation? Describe the steps in simulation study. 2

d) In the bombing problems, the ammunition depot area is a circle of 5000m radius. The point of impact is normally distributed around the aiming point with standard deviation of 500m in X direction and 300m in the Y direction. Simulate the bombing operation for 20 strike and find the percentage of strikes on target. Use Normal Random Number: 5

For X direction - [.23, -1.16, 0.39, -1.90, -0.78, -0.02, -0.40, -0.66, 1.14, 0.07, 0.53, 0.03, 1.19,0.11, 0.16, -0.82, 0.42, -0.89, 0.49, -0.21]

For Y direction- [0.24, -0.02, 0.64, -1.04, 0.68, -0.47, -0.75, -0.44, 1.21, -0.08, -1.56, 1.49, -1.19, -1.86, 1.21, 0.75, -1.50, 0.19, -1.44, 0.65]

2. a. Let us consider a case where two chemicals ch_1 and ch_2 react to produce a third chemical ch_3. The rate of reaction will depend upon a large number of factors like the ratio in which ch_1 and ch_2 are mixed, the temperature, the pressure and the humidity etc. In addition to the forward reaction where ch_1 and ch_2 react to produce ch_3, there may also be backward reaction, where the ch_3 decomposes back into ch_1 and ch_2. Let the rate of formation of ch_3 be proportional to the product of the amount of ch_1 and ch_2 present in the mixture and let the rate of decomposition of ch_3 be proportional to its amount in the mixture. If c_1,c_2 and c_3 are the amount of ch_1, ch_2 and ch_3 at any instant of time t, the rate of change of c_1,c_2 and c_3 are given by the following differential equations: 07

$$\frac{dc_1}{dt} = k_2c_3 - k_1c_1c_2$$

$$\frac{dc_2}{dt} = k_2c_3 - k_1c_1c_2$$

$$\frac{dc_3}{dt} = 2k_1c_1c_2 - 2k_2c_3$$

Where k_1 *and* k_2 are constants. If the temperature, pressure and humidity are constant and have no effect on the rate of reaction and $k_1 = 0.025, k_2 = 0.01, c_1[0] = 50.00, c_2[0] = 25.00, c_3[0] = 0.00$ *and* $\Delta_t = 1\ sec$; simulate the chemical reaction for 20 sec.

b. In a pure pursuit, there is a target bomber which moves along a predetermine path (given on table below) and there is a pursuer aircraft who follows the target bombers, redirecting itself towards the target bombers at fixed intervals of time.

Target bombers moving path:

Time,t	0	1	2	3	4	5	6	7	8
xb(t)	100	100	120	129	140	149	158	168	179
yb(t)	0	3	6	10	15	20	26	32	37
Time,t	9	10	11	12	13	14	15		
xb(t)	188	198	209	219	226	234	240		
yb(t)	34	30	27	23	19	16	14		

The fighter aircraft following the enemy bombers with speed at 25KM/min to destroy it. The fighter aircraft is at position *xf,yf*(0, 60) when it sights the bomber that is at time t=0, the time at which the pursuit begins. The fighter corrects its direction after a fixed interval of one minute, so as to point towards a bomber and shoots the bomber by firing a missile as soon as it is within a distance of 12 KM. The pursuit ends. In case the bomber is not shot within 15 minutes of the pursuit, the pursuit is abandoned. Simulate the problem to determine the pursuit is end/abandoned.

3. a) When the mid square random number generators produce unexpected result? Write down the qualities of an efficient random number generator. 04

b) The following sequence of random numbers have been generated 0.037, 0.55, 0.71, 0.97, 0.63, 0.29, 0.84, 0.78, 0.23, 0.17. Use Kolmogorov-Smirnov test with $\alpha = 0.05$ to determine, If these numbers are uniformly distributed over the interval 0 to 1. Note that for $\alpha = 0.05$ and N=10 the critical value is 0.410. 05

c) ~~Write down the qualities of an efficient random number generator.~~ Use the multiplicative congruential method to generate a sequence of 10 three-digit random numbers. Let, $r_0 = 117, a = 3, m = 1000$. 05

4. a) Why it is needed to test the randomness of pseudo random numbers? Write down the different types of testing system of randomness. 02

b) A sequence of 10,000 five digit random numbers has been generated, and an analysis of numbers indicate that there 3075 numbers having five different digits, 4935 having a pair, 1135 having two pairs, 695 having three of a kind, 105 having full house, 54 having four of a kind and one having all five of a kind. Use Poker test to determine if these random numbers are independent at $\alpha = 0.01$. Note that for $\alpha = 0.01$ and N=06 the critical value is 16.8. 05

c) The two-digit random numbers generated by a random variable are given below. Determine Chi-Square. Is it acceptable at 95% confidence level? [Use Chi-test table is at page 4]

21	81	92	28	96	20	68	52	79	84
82	62	12	08	92	83	74	85	60	49
48	37	65	74	22	11	28	10	55	82
72	95	08	85	79	95	86	11	16	52
70	55	50	87	67	51	72	38	29	62
71	12	07	75	56	34	40	87	24	86
18	82	41	29	63	06	84	01	20	06
06	33	14	79	25	65	57	47	74	68
54	38	81	07	88	96	70	85	29	13
12	91	26	57	30	22	90	03	13	31

5. a) What is stochastic process? 02

b) A pair of dice is rolled once. The sum of the two numbers on the dice represents the outcome for a random variable x. Determine the probability distribution of x. What is the probability that x is even? 04

c) Why is Gaussian distribution called bell curve? (mean, mode, median all are same) 02

d) Generate five random observations from Normal distribution with mean =5 and standard deviation 4. 04

e) Give some applications of Poisson distribution. 02

6. a) Define the following terminologies: (i) Arrival rate (ii) Service rate (iii) Reneging. 06

b) Explain and illustrate the cost of customer waiting time and the idle capacity. 03

c) A factory has a large number of semiautomatic machines. On 50%of the working days none of the machine fail. On 30% of the days one machine fails and on 20% of the days two machines fail. The maintenance staff on the average puts 65% of the machines in order in one day, 30% in two days, and the remaining 5% in three days. Estimate the average length of queue, average waiting time, and the server loading that is the fraction of time for which server is busy. 03

d) According to two servers in parallel queuing system, illustrate a mixture of components A and B pass through workstation I. 02

Values of the Chi-Squared Distribution

df	Probability of the Chi-Square								
	0.995	0.975	0.9	0.5	0.1	0.05	0.05	0.01	0.005
1	0.000	0.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	0.236	11.070	12.832	15.086	16.750
6	0.676	1.237	2.402	5.348	10.645	12.592	14.449	16.812	18.548
7	0.989	1.690	2.833	6.346	12.017	14.067	16.013	18.475	20.278
8	1.344	2.180	3.490	7.344	13.362	15.507	17.535	20.090	21.955
9	1.735	2.700	4.168	8.343	14.684	16.919	19.023	21.666	23.589
10	2.156	3.247	4.865	9.342	15.987	18.307	20.483	23.209	25.188

Patuakhali Science and Technology University

6th Semester (L-3, S-II), Final Exam of B. Sc. Engg. (CSE), July -Dec. 2018

Course Code: CIT 323 Course Title: Simulation and Modeling

Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 of the following questions:

Simulation

1. (a). Discuss about the component used in next event time-advance approach. 4

(b). Show the comparison between

i) analytical solution and simulation

ii) static simulation and dynamic simulation. 4

(c). For a single server queuing system, inter-arrival times of customer are 0.2, 0.8, 0.4, 2.0, 1.0, 0.2, 0.5, 1.0 and service times are 1.5, 1.0, 0.6, 1.2, and 0.6. Show the snapshot of computer representation at time 0 and at each of the 6 succeeding event times. 6

2. (a). What are differences between next event time advance approach and fixed increment time advance approach? 3

(b). What are the limitations of mid-square method? Evaluate the performance of Mid-square method if seeds are 6785, 1379, 1357. 7

(c). Generate a sequence of 15 random numbers between 20 to 30 such that $r_{n+1} = (7r_n + 15) \bmod m$, where $r_0 = 11$. m = prime number 4

3. (a). Perform the Kolmogorov-Smirnov test to testify the uniformity of following random numbers at 95% level of significance. 4

0.37, 0.1, 0.5, 0.88, 0.37, 0.29, 0.84, 0.78, 0.08, 0.46

At $\alpha = 0.01$ and $N = 10$, critical values is 0.368; At $\alpha = 0.05$ and $N = 10$, critical values is 0.410

(b). Generate two digit 40 random numbers between 0 to 50 by a multiplicative congruential method. Apply Chi-Squared test to verify the randomness. Where, the accepted value of Chi-square for four degree of freedom at 95% confidence level is 9.448. 6

(c) What do you mean by system simulation? Write down the properties of random numbers. 4

4. (a) Define computer simulation models. Describe the steps in simulation study. 2

(b) Define distributed simulation. Describe the iterative process of verification and validation of simulation models. 4

(c) A drunkard moved from a point to a destination and takes step in four directions, backward, forward, right, and left. The probabilities associated with these are 10%, 50%, 15% and 25% respectively. The distances covered to backward, forward, right, and left are 30 cm, 80 cm, 60 cm and 40 cm respectively. Simulate the walk for 20 steps and find the location at end of 20 steps while starting point is (0, 0) on the X-Y scales. [USE Random Number 1 to 20 from table] 8

5. (a) Write down the difference between monte-carlo and stochastic simulation. 2

(b) An ammunition depot, of circular shape measuring 500 m of radius is under attack by a squadron of bombers. In each sortie 10 bombers drop bombs, one each, on the ammunition depot. If the bomb lands anywhere in the marked area, it is a hit, otherwise it is a miss. All the bombers aim at the center of the area. The point of strike is assumed to be normally distributed around the aiming point with a standard deviation of 500 m in x-direction and 300m in the y-direction. Simulate operation of bombing operation for 20 strikes and determine the percentage number of strikes, Which are on the target. [USE Random Number 21 to 40 for X coordinate and Random Number 41 to 60 for Y coordinate ,from table] 8

(c) Define Monte-Carlo method. Solve the integrals $I = \int_2^5 x^3 dx$ by monte-carlo method. [USE Random Number 61 to 70 for X coordinate and Random Number 71 to 80 for Y coordinate, from table] 4

(a) Define critical path. Describe the benefits of using simulation of PERT and CPM for planning, scheduling and controlling of a large and complex software project. 5

(b) In a pure pursuit, there is a target bombers which moves along a predetermine path (given on table below) and there is a pursuer aircraft who follows the target bombers, redirecting itself towards the target bombers at fixed intervals of time. 6

Target bombers moving path:

Time,t	0	1	2	3	4	5	6	7	8
xb(t)	160	161	167	172	181		199	215	239
yb(t)	0		9	11	14	24	34	41	52

Time,t	9	10	11	12	13	14	15		
xb(t)	245	254	265	271	289	293	300		
yb(t)	45	41	36	39	34	28	12		

The fighter aircraft following the enemy bombers with speed at 30KM/min to destroy it. The fighter aircraft is at position *xf, yf* (50, 50) when it sights the bomber that is at time t=0, the time at which the pursuit begins. The fighter corrects its direction after a fixed interval of one minute, so as to point towards a bomber and shoots the bomber by firing a missile as soon as it is within a distance of 10 KM. The pursuit ends. In case the bomber is not shot within 15 minutes of the pursuit, the pursuit is abandoned. Simulate the problem to determine the pursuit is end/abandoned.

(c) Consider a chemical reaction of two gas H_2 *and* O_2 react to produce H_2O. The rate of the reaction is depending upon the ratio in which H_2 *and* O_2 are mixed and some other factor like temperature, pressure and humidity which remains constant during the reaction. There will also happen backward reaction where H_2O decomposes back into H_2 *and* O_2. If C_1, C_2 *and* C_3 are the amount of H_2, O_2 *and* H_2O at any instant of time t, the rate of increase of H_2, O_2 *and* H_2O are given by the following differential equations: 5

$$\frac{dC_1}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_2}{dt} = K_2C_3 - K_1C_1C_2$$

$$\frac{dC_3}{dt} = 2K_1C_1C_2 - 2K_2C_3$$

Where, K_1 *and* K_2 are constants. If K_1=0.025, $K_2 = 0.01$ and at time t=0, $C_1 = 50.00\ m^3$, $C_2 = 25.00\ m^3$, $C_3 = 0.00\ m^3$, $\Delta t = 0.2\ minute$. Simulate the experiment for 3 minute to determine the amount of H_2, O_2 *and* H_2O.

Patuakhali Science and Technology University (PSTU)

Department of Computer Science and Communication Engineering (CSE)

Final Examination (LAB), July-December: 2019

Program: B. Sc. Engg.(CSE)

Session: 2016-17

Course Code : CIT-323	Course Title : Simulation and Modeling Sessional
Full Time: 3 hours	Full Marks: 70

Answer the marked questions

Part A (Open Source ; Hard copy only)

1.	Simulation of a pure pursuit problem	40
2.	Simulation of exterior ballistics	40
3.	Perform chi-squared test of uniformity and auto correlation on two-digit random number generated by multiplicative congruential method.	40
4.	Simulate an industrial reliability problem	40
5.	Simulate a counter service	40
6.	Simulate a profit analysis problem	40
7.	Simulate an inventory problem	40
8.	Simulate a manufacturing system	40

Part B

9	Generate five random observations from each of the following distributions. a. Exponential with mean=5 b. Erlang with mean=5 and k=3 c. Norman with mean=5 and standard deviation=4	10
10	Continuous Assessments in Laboratory	10
13	Viva-Voce	10

CamScanner

Random Number (RN) Table

(N.B. : Pseudo Random value as test data)

S.N.	RN	S.N.	RN	S.N.	RN	S.N.	RN
1.	67	32.	.003	63.	18		
2.	23	33.	-1.19	64.	45		
3.	01	34.	0.11	65.	15		
4.	95	35.	0.16	66.	27		
5.	84	36.	-0.82	67.	48		
6.	56	37.	0.42	68.	43		
7.	77	38.	-0.89	69.	40		
8.	76	39.	0.49	70.	47		
9.	18	40.	-0.21	71.	.57		
10.	82	41.	0.24	72.	.18		
11.	45	42.	-0.02	73.	.00		
12.	55	43.	0.64	74.	.90		
13.	29	44.	-1.04	75.	.05		
14.	68	45.	0.68	76..	.77		
15.	21	46.	-0.47	77.	.66		
16.	15	47.	-0.75	78.	.10		
17.	86	48.	-0.44	79.	.76		
18.	00	49.	1.21	80.	.42		
19.	89	50.	-0.08				
20.	43	51.	-1.56				
21.	.23	52.	1.49				
22.	-1.16	53.	-1.19				
23.	0.39	54.	-1.86				
24.	-1.90	55.	1.21				
25.	-0.78	56.	0.75				
26.	-0.02	57.	-1.50				
27.	-0.04	58.	0.19				
28.	-0.66	59.	-1.44				
29.	1.41	60.	0.65				
30.	0.07	61.	22				
31.	0.53	62.	25				

Patuakhali Science and Technology University

Department of Computer Science Information Technology(CSIT)

Midterm Examination, July-December-2021

Program: B. Sc. Engg.(CSE)

Session : **2018-19**

Course Code : CIT-324

Course Title : Simulation and Modeling Sessional

Full Time: 1 Hours

Full Marks: 15

Answer the marked question

1. Calculate the area of an irregular figure by Monte Carlo Method. 7.5
2. Simulate the Gambling Game by Monte Carlo Method and analyze profit-loss. 7.5
3. Find the value of a numerical integration by Monte Carlo Method of any differential equation. 7.5
4. Determination of the value of π by Monte Carlo Method. 7.5
5. Solve Random walk by Monte Carlo Method. 7.5
6. Simulate Reliability Problem. 7.5
7. Implement Kolmogorov test with midsquare random numbers. 7.5
8. Implement Chi-Square test with multiplicative congruential random numbers. 7.5
9. Implement chi-square test for autocorrelation with additive congruential random numbers. 7.5
10. Implement poker test with multiplicative congruential random numbers. 7.5

0 1 2 3 4 5 6 7 8 9

H T

100

50

40

70

74

R L 75-99

Random Number (RN) Table

(N.B. : Pseudo Random value as test data) 350

S.N.	RN	S.N.	RN	S.N.	RN	S.N.	RN
1.	67	21.	.23	41.	0.24	61.	22
2.	23	22.	-1.16	42.	-0.02	62.	25
3.	01	23.	0.39	43.	0.64	63.	28
4.	95	24.	-1.90	44.	-1.04	64.	45
5.	84	25.	-0.78	45.	0.68	65.	~~15~~ 25
6.	56	26.	-0.02	46.	-0.47	66.	27
7.	77	27.	-0.04	47.	-0.75	67.	48
8.	76	28.	-0.66	48.	-0.44	68.	43
9.	18	29.	1.41	49.	1.21	69.	40
10.	82	30.	0.07	50.	-0.08	70.	47
11.	45	31.	0.53	51.	-1.56	71.	.57
12.	55	32.	.003	52.	1.49	72.	.18
13.	29	33.	-1.19	53.	-1.19	73.	.00
14.	68	34.	0.11	54.	-1.86	74.	.90
15.	21	35.	0.16	55.	1.21	75.	.05
16.	15	36.	-0.82	56.	0.75	76..	.77
17.	86	37.	0.42	57.	-1.50	77.	.66
18.	00	38.	-0.89	58.	0.19	78.	.10
19.	89	39.	0.49	59.	-1.44	79.	.76
20.	43	40.	-0.21	60.	0.65	80.	.42